Project Title : Online Pharmacy Management System

Subject : Database Management System (DBMS)

Name : Abhishek Yadav

Roll.no : 2410030746

Submitted To : Mr. Aditya Bangia

Problem Statement

Managing medicine stock manually in medical stores or online pharmacies is difficult and time-consuming. Tracking expiry dates, monitoring low stock levels, maintaining supplier details, and recording sales transactions can lead to errors if not handled properly.

The objective of this project is to design a database system that:

- Manages medicine stock efficiently
- Identifies expired medicines
- Provides low stock alerts
- Maintains accurate sales records
- Organizes supplier and customer information

This system ensures data accuracy, consistency, and security.

Database Schema (Tables)

Customer Table

Column Name	Data Type	Constraint
customer_id	INT	Primary Key
customer_name	VARCHAR(50)	NOT NULL
phone	VARCHAR(15)	

Supplier Table

Column Name	Data Type	Constraint
supplier_id	INT	Primary Key
supplier_name	VARCHAR(50)	NOT NULL
supplier_address	VARCHAR(100)	

Medicine Table

Column Name	Data Type	Constraint
medicine_id	INT	Primary Key
medicine_name	VARCHAR(50)	NOT NULL
price	DECIMAL(10,2)	NOT NULL
stock	INT	DEFAULT 0
supplier_id	INT	Foreign Key → Supplier(supplier_id)

Sales Table

Column Name	Data Type	Constraint
sale_id	INT	Primary Key
customer_id	INT	Foreign Key → Customer(customer_id)
medicine_id	INT	Foreign Key → Medicine(medicine_id)
quantity	INT	NOT NULL
sale_date	DATE	DEFAULT CURRENT_DATE

SQL QUERIES

DML OPERATION –

1. INSERT OPERATION –

INSERT INTO Supplier VALUES

(1,'ABC Pharma','9876543210','Delhi'),

(2,'MediCare Ltd','9123456780','Mumbai'),

(3,'HealthPlus','9012345678','Noida'),

(4,'LifeCare Pharma','9988776655','Ghaziabad'),

(5,'Wellness Drugs','8899776655','Pune');

```
MariaDB [pharmacy]> INSERT INTO Supplier VALUES
    -> (1,'ABC Pharma','9876543210','Delhi'),
    -> (2,'MediCare Ltd','9123456780','Mumbai'),
    -> (3,'HealthPlus','9012345678','Noida'),
    -> (4,'LifeCare Pharma','9988776655','Ghaziabad'),
    -> (5,'Wellness Drugs','8899776655','Pune');
Query OK, 5 rows affected (0.086 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

INSERT INTO Medicine VALUES

(101,'Paracetamol','Tablet',20.00,50,'2026-05-01',1),

(102,'Amoxicillin','Capsule',35.00,8,'2025-02-15',2),

(103,'Cough Syrup','Liquid',60.00,15,'2025-12-10',3),

(104,'Vitamin D','Supplement',80.00,5,'2024-11-20',1),

(105,'Ibuprofen','Tablet',25.00,30,'2026-01-10',2),

(106,'Azithromycin','Tablet',45.00,12,'2025-09-18',3),

(107,'Insulin','Injection',150.00,20,'2025-07-25',4),

(108,'Antacid','Liquid',55.00,18,'2026-03-05',5),

(109,'Calcium','Supplement',70.00,9,'2025-08-30',4),

(110,'ORS','Powder',15.00,40,'2026-06-12',5);

```
MariaDB [pharmacy]>
MariaDB [pharmacy]> INSERT INTO Medicine VALUES
    -> (101,'Paracetamol','Tablet',20.00,50,'2026-05-01',1),
    -> (102,'Amoxicillin','Capsule',35.00,8,'2025-02-15',2),
    -> (103,'Cough Syrup','Liquid',60.00,15,'2025-12-10',3),
    -> (104,'Vitamin D','Supplement',80.00,5,'2024-11-20',1),
    -> (105,'Ibuprofen','Tablet',25.00,30,'2026-01-10',2),
    -> (106,'Azithromycin','Tablet',45.00,12,'2025-09-18',3),
    -> (107,'Insulin','Injection',150.00,20,'2025-07-25',4),
    -> (108,'Antacid','Liquid',55.00,18,'2026-03-05',5),
    -> (109,'Calcium','Supplement',70.00,9,'2025-08-30',4),
    -> (110,'ORS','Powder',15.00,40,'2026-06-12',5);
Query OK, 10 rows affected (0.012 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

INSERT INTO Customer VALUES

(1,'Rahul','9998887776'),

(2,'Anita','8887776665'),

(3,'Amit','7776665554'),

(4,'Sneha','7666555443'),

(5,'Rohit','7555444332'),

(6,'Neha','7444333221');

```
MariaDB [pharmacy]> INSERT INTO Customer VALUES
    -> (1,'Rahul','9998887776'),
    -> (2,'Anita','8887776665'),
    -> (3,'Amit','7776665554'),
    -> (4,'Sneha','7666555443'),
    -> (5,'Rohit','7555444332'),
    -> (6,'Neha','7444333221');
Query OK, 6 rows affected (0.014 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

INSERT INTO Customer VALUES (7,'Karan','7333222111');

```
MariaDB [pharmacy]> INSERT INTO Customer VALUES (7,'Karan','7333222111');
Query OK, 1 row affected (0.015 sec)

MariaDB [pharmacy]> SELECT * FROM Customer;
+-------------+-------+------------+
| customer_id | name  | phone      |
+-------------+-------+------------+
|           1 | Rahul | 9998887776 |
|           2 | Anita | 8887776665 |
|           3 | Amit  | 7776665554 |
|           4 | Sneha | 7666555443 |
|           5 | Rohit | 7555444332 |
|           6 | Neha  | 7444333221 |
|           7 | Karan | 7333222111 |
+-------------+-------+------------+
7 rows in set (0.001 sec)
```

INSERT INTO Sales VALUES

(1,101,1,2,'2026-02-01'),

(2,102,2,1,'2026-02-05'),

(3,103,3,3,'2026-02-10'),

(4,104,4,1,'2026-02-12'),

(5,105,5,2,'2026-02-15'),

(6,106,6,1,'2026-02-18'),

(7,107,1,1,'2026-02-20'),

(8,108,2,2,'2026-02-22'),

(9,109,3,1,'2026-02-25'),

(10,110,4,3,'2026-02-28');

```
MariaDB [pharmacy]> INSERT INTO Sales VALUES
    -> (1,101,1,2,'2026-02-01'),
    -> (2,102,2,1,'2026-02-05'),
    -> (3,103,3,3,'2026-02-10'),
    -> (4,104,4,1,'2026-02-12'),
    -> (5,105,5,2,'2026-02-15'),
    -> (6,106,6,1,'2026-02-18'),
    -> (7,107,1,1,'2026-02-20'),
    -> (8,108,2,2,'2026-02-22'),
    -> (9,109,3,1,'2026-02-25'),
    -> (10,110,4,3,'2026-02-28');
Query OK, 10 rows affected (0.020 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

UPDATE OPERATION –

Before update:

SELECT * FROM Medicine WHERE medicine_id = 101;

```
MariaDB [pharmacy]> SELECT * FROM Medicine WHERE medicine_id = 101;
+-------------+-------------+----------+-------+-------+-------------+-------------+
| medicine_id | name        | category | price | stock | expiry_date | supplier_id |
+-------------+-------------+----------+-------+-------+-------------+-------------+
|         101 | Paracetamol | Tablet   | 20.00 |    50 | 2026-05-01  |           1 |
+-------------+-------------+----------+-------+-------+-------------+-------------+
1 row in set (0.001 sec)
```

UPDATE Medicine

SET price = 30

WHERE medicine_id = 101;

```
MariaDB [pharmacy]> UPDATE Medicine
    -> SET price = 30
    -> WHERE medicine_id = 101;
Query OK, 1 row affected (0.005 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [pharmacy]> SELECT * FROM Medicine WHERE medicine_id = 101;
+-------------+------------+----------+-------+-------+-------------+-------------+
| medicine_id | name       | category | price | stock | expiry_date | supplier_id |
+-------------+------------+----------+-------+-------+-------------+-------------+
|         101 | Paracetamol | Tablet   | 30.00 |    50 | 2026-05-01  |           1 |
+-------------+------------+----------+-------+-------+-------------+-------------+
1 row in set (0.001 sec)
```

3. DELETE OPERATION –

DELETE FROM Customer WHERE Customer_id = 7;

```
MariaDB [pharmacy]> DELETE FROM Customer
    -> WHERE customer_id = 7;
Query OK, 1 row affected (0.013 sec)

MariaDB [pharmacy]> SELECT * FROM Customer;
+-------------+-------+------------+
| customer_id | name  | phone      |
+-------------+-------+------------+
|           1 | Rahul | 9998887776 |
|           2 | Anita | 8887776665 |
|           3 | Amit  | 7776665554 |
|           4 | Sneha | 7666555443 |
|           5 | Rohit | 7555444332 |
|           6 | Neha  | 7444333221 |
+-------------+-------+------------+
6 rows in set (0.001 sec)
```

Query 1- Select With Condition

SELECT * FROM Medicine

WHERE stock < 10;

```
MariaDB [pharmacy]> SELECT * FROM medicine
    -> WHERE stock < 10;
+-------------+-------------+------------+-------+-------+-------------+-------------+
| medicine_id | name        | category   | price | stock | expiry_date | supplier_id |
+-------------+-------------+------------+-------+-------+-------------+-------------+
|         102 | Amoxicillin | Capsule    | 35.00 |     8 | 2025-02-15  |           2 |
|         104 | Vitamin D   | Supplement | 80.00 |     5 | 2024-11-20  |           1 |
|         109 | Calcium     | Supplement | 70.00 |     9 | 2025-08-30  |           4 |
+-------------+-------------+------------+-------+-------+-------------+-------------+
3 rows in set (0.001 sec)
```

Query 2- Join Query

SELECT c.name, m.name, s.quantity

FROM Sales s

JOIN Customer c ON s.customer_id = c.customer_id

JOIN Medicine m ON s.medicine_id = m.medicine_id;

```
MariaDB [pharmacy]> SELECT c.name, m.name, s.quantity
    -> FROM Sales s
    -> JOIN Customer c ON s.customer_id = c.customer_id
    -> JOIN Medicine m ON s.medicine_id = m.medicine_id;
+-------+--------------+----------+
| name  | name         | quantity |
+-------+--------------+----------+
| Rahul | Paracetamol  |        2 |
| Anita | Amoxicillin  |        1 |
| Amit  | Cough Syrup  |        3 |
| Sneha | Vitamin D    |        1 |
| Rohit | Ibuprofen    |        2 |
| Neha  | Azithromycin |        1 |
| Rahul | Insulin      |        1 |
| Anita | Antacid      |        2 |
| Amit  | Calcium      |        1 |
| Sneha | ORS          |        3 |
+-------+--------------+----------+
10 rows in set (0.001 sec)
```

Query 3- Aggregate Function

SELECT m.name, SUM(s.quantity) AS total_sold

FROM Sales s

JOIN Medicine m ON s.medicine_id = m.medicine_id

GROUP BY m.name;

```
MariaDB [pharmacy]> SELECT m.name, SUM(s.quantity) AS total_sold
    -> FROM Sales s
    -> JOIN Medicine m ON s.medicine_id = m.medicine_id
    -> GROUP BY m.name;
+--------------+------------+
| name         | total_sold |
+--------------+------------+
| Amoxicillin  |          1 |
| Antacid      |          2 |
| Azithromycin |          1 |
| Calcium      |          1 |
| Cough Syrup  |          3 |
| Ibuprofen    |          2 |
| Insulin      |          1 |
| ORS          |          3 |
| Paracetamol  |          2 |
| Vitamin D    |          1 |
+--------------+------------+
10 rows in set (0.001 sec)
```

Query 4- Subquery

SELECT name FROM Medicine

WHERE price > (SELECT AVG(price) FROM Medicine);

```
MariaDB [pharmacy]> SELECT name FROM Medicine
    -> WHERE price > (SELECT AVG(price) FROM Medicine);
+-------------+
| name        |
+-------------+
| Cough Syrup |
| Vitamin D   |
| Insulin     |
| Calcium     |
+-------------+
4 rows in set (0.001 sec)
```

Conclusion –

This project helps in managing medicine stock, suppliers, customers, and sales using a database system. It makes data handling easy, reduces errors, and allows quick access to information using SQL queries. Overall, it improves efficiency and organization in pharmacy management.